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Republic of Zambia

Ministry of Education

SCIENCE SYLLABUS
GRADE 4 - 6

DEVELOPED BY THE CURRICULUM DEVELOPMENT CENTRE
LUSAKA
2024

Republic of Zambia

MINISTRY OF EDUCATION

SCIENCE SYLLABUS
GRADE 4-6

Developed by The Curriculum Development Centre

2024

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VISION

Quality, life long education for all which is accessible, inclusive and relevant to individual, national and global needs and value systems

PREFACE

The **Upper Primary Science Syllabus** has been reviewed by the need to provide a national age-appropriate curriculum for learners at Grade 4 to 6. The syllabus aims to provide quality education that is aligned with the Competence-Based Curriculum and 21st Century Skills. It also aims to impart the knowledge, skills, values and positive attitudes that are aimed at enabling learners to live and grow into productive and useful members of their communities and the Zambian society.

This syllabus comprises four branches of science namely; Agricultural Science, Biology, Chemistry and Physics.

This learning area is designed to promote and support the linkage with the lower primary education and also act as a foundation for secondary education.

The topics provided in this syllabus are aimed at facilitating hands-on experiences through manipulation of real objects, interaction with nature and learning through research and practical work. Thus, the learners will develop knowledge, desirable lifelong skills, values and positive attitudes needed for their personal and national development.

Cross cutting themes and emerging issues such as national values and principles, life skills and health Education, climate change and financial education have been incorporated in the syllabi to ensure that learners cultivate a mindset, skills, values and positive attitudes that prepare them to live responsible lives and be protected from life threatening vices.

It is hoped that the syllabus will make learning at upper primary school level more meaningful and enjoyable as it is highly activity oriented and allows for a smooth transition from primary to secondary school.

Joel Kamoko (Mr)
Permanent Secretary- Education Services

MINISTRY OF EDUCATION

ACKNOWLEDGMENT

This syllabus is designed to provide the topics necessary to be offered in science at primary school level. It is intended to guide teachers and other experts in the field to appropriately offer relevant lessons in science at this level.

My gratitude goes to individuals, institutions and organisations that provided the financial and technical input to the successful development of this syllabi. These include; the teachers, lecturers from colleges and public universities in Zambia and officers from the Directorates of National Science Centre, and Primary Education in the Ministry of Education.

Finally, I recognise the commitment and hard work of all the staff at the Curriculum Development Centre in ensuring that this syllabus comes to reality.

Charles Ndakala, (Dr.)
Director – Curriculum Development
MINISTRY OF EDUCATION

INTRODUCTION

The **Upper Primary Science syllabus** is designed for learners pursuing upper primary school level in Zambian schools. The course will provide an engaging and explorative journey designed to spark curiosity, nurture foundational skills, and instill a love for learning in the young minds. In this course, there is recognition of the immense potential within each learner to strive to create a dynamic and stimulating environment where they can joyfully discover the wonders of +science and based on the belief that children are natural-born scientists. From the moment they begin to explore the world around them, they engage in a process of observation, questioning, and experimentation. Therefore, this syllabus aims to harness and enhance these innate abilities thereby providing a solid foundation for Science Technology Engineering and Mathematics (STEM) education and smooth transition to secondary education.
The course covers the following topics:

1. The Human Body
2. Nutrition and Health
3. The Environment
4. Plants and Animals
5. Materials and Energy

Objectives

The main objective of providing Science education to every learner is to;

- Foster curiosity to nurture a sense of wonder by encouraging young learners to ask questions, explore their surroundings, and make connections between the scientific concepts they encounter.
- Foster scientific inquiry that promotes a hands-on approach to science by engaging young learners in simple, age-appropriate experiments to observe cause and effect, explore the properties of materials, and develop a keen interest in the world of science.
- Encourage collaborative learning by recognising the importance of social interactions in the learning process. The activities should be designed to promote teamwork, communication, and the sharing of ideas, creating a supportive community of learners.
- Encourage sustainable environmental management practices to evade the impact of climate change.

Structure of the Syllabus

The syllabus contains five concepts which run as topics. These are;

the Human Body, Nutrition and Health, The Environment, Plants and Animals, and Materials and Energy. The concepts have been outlined in such a manner that they are repetitive at each grade level. As much as the concepts are repeating across the grade levels, the content, however, gets complex. The syllabus outlines the content in the form of topics, sub-topics while the specific competences are meant to respond to the realisation of scientific concepts.

The topics, sub-topics and competences are arranged in such a manner that makes it easy for quick reference. While certain topics may appear similar across different grade levels, the requisite depth of understanding, skill acquisition, and value internalisation varies significantly. To optimise the learning process, teachers must design pedagogical approaches that leverage existing knowledge as a foundation for subsequent instruction.

Suggested Teaching Methodology

The approach to teaching and learning in this syllabus is anchored on learner-centred teaching methods. This is best achieved when learners are actively involved in the learning process through hands- on activities. However, each individual learner has their own unique needs such as pace of learning, experiences in life and abilities. To accommodate this, the teacher must determine the needs of each individual learner, and shape the learning experiences accordingly.

The hands-on learning enhances creativity, analysis, problem solving and an investigative approach which can be taught effectively using a variety of methods both in the classroom and outside. Therefore, teaching methods must be varied, but flexible within well-structured sequences of lessons. These include, but not limited to conducting experiments, study tours, field work, presentations, practical work, inquiry learning and project work. In order to foster collaboration, learners are encouraged to work in pairs, small groups as well as recognising individual efforts.

Therefore, in order to develop learners with skills and values that can contribute to the development of society, the starting point for teaching and learning is the experience gained from the family, community and through interaction with the environment. Thus, learning in school must build on the learner's prior knowledge and experience.

Time Allocation

The standard minimum teacher - learner contact time for Science at Upper Primary school level (Grades 4 to 6) is 4 hours per week, translating into six (6) periods. The duration for a single period is 40 minutes.

Assessment

Assessment is a vital component of the teaching and learning process. It provides valuable insights into learners' abilities to develop the desired competences. To effectively assess learners' acquisition of competences a variety of methods will be employed, including exercises, observations, interviews, checklists, tests, projects, and examinations.

School-Based Assessment (SBA) is a key strategy for continuous assessment. It involves the ongoing monitoring of learner progress through various activities and tasks. Each learner will maintain a portfolio to document their work and achievements. The Examinations Council of Zambia (ECZ) will provide detailed guidelines to ensure that SBA is conducted appropriately by teachers and that assessment results are managed effectively.
By incorporating a diverse range of assessment methods and implementing a comprehensive SBA system, educators will gain a comprehensive understanding of the learners learning.

Key Competences to be developed

S/N	COMPETENCE	DESCRIPTORS
1	Analytical Thinking	• Identify patterns • Compile data, create mental images and address issues • Evaluate solutions
2	Communication	• Use mathematical/scientific language in different situations • Express oneself using different media and symbols • Ask for feedback
3	Creativity and Innovation	• Explore the objects around them. • Show creativity and innovativeness • Explore areas of interest
4	Collaboration	• Solving puzzle in groups • Play with peers to build relationships • Participate in and express themselves through play activities
5	Critical Thinking	• Ask and answer simple questions • Classify objects according to their attributes • Manipulate different objects • Solve simple problems in life • Match different things according attributes • Arrange objects according to attributes • Compare similarities or differences between objects • Explore the environment • Differentiate good from bad • Recognize and name items in the environment
6	Environmental Sustainability	• Dispose trash in the designated place. • Adhere to best practices in environmental management. • Identify a clean environment. • Identify types of waste in local environment
8	Problem Solving	• Make connections/link with the inner world or social environment • Use numeracy patterns and relations to solve problems • Manipulate numbers, shapes and symbols to complete a task

GRADE 4

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTEDSTANDARD
4.1 THE HUMAN BODY	**4.1.1 The Respiratory System**	4.1.1.1 Demonstrate understanding of the respiratory system in the human body	• Describing the respiratory system in humans • Analysing the main parts of the respiratory system in man (*nose, trachea, bronchi, and lungs*) • Drawing of the human respiratory system • Making models and charts of the human respiratory system illustrating the movement of air from outside into the lungs and vice-versa • Investigating respiratory diseases and disorders in man (*Bronchitis, Tuberculosis, Influenza, Asthma…)* • Creating a self-care plan to promote respiratory health	Understanding of the respiratory system demonstrated sa tisfactorily

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTEDSTANDARD
	4.1.2 Blood Circulatory System	**4.1.2.1** Demonstrate understanding of the blood circulatory system in the human body	• Describing the Blood Circulatory System in man • Analysing the components of blood (*red blood cells, white blood cells, platelets and plasma*) • Drawing and labeling the parts of the human heart collaboratively • Making a simple longitudinal model of the human blood circulating system illustrating the flow of blood around the body • Making a simple longitudinal model of the human heart • Tracking the flow of blood in the vessels (*veins and arteries*) • Analysing the importance of the Blood Circulatory System in man • Calculating the pulse rate • Investigating common cardiac diseases in the community (*Heart failure, coronary heart disease …)* • Analysing the possible causes of cardiac diseases (*Unhealthy diets, tobacco use, alcohol abuse, diabetes, high blood pressure…)*	• Understanding of the blood circulatory system demonstrated appropriately

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTEDSTANDARD
4.2 NUTRITION AND HEALTH	**4.2.1 Classification of Food**	**4.2.1.1** Classify foods based on their nutritional content	• Collecting different foods found in the local environment (*raw and processed foods*) • Sorting a variety of foods into their respective groups based on the nutritive content (*cereals and tubers; fish, insects and meat; pulses, beans, nuts and seeds; dairy products; vegetables; and fruits)* • Classifying food nutrients into proteins, carbohydrates, fats, vitamins and minerals	• Foods classified into their respective groups correctly
	4.2.2 Healthy Diets	**4.2.2.1** Practise eating healthy diets	• Designing healthy diets for persons with different dietary needs • Discussing the importance of eating smaller quantities from all the food groups regularly (Apportionment) • Making informed choices about healthy living (*taking light exercises, taking the Body Mass Index (BMI) test, taking enough water daily…*)	• Eating healthy diets practised accordingly

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
	4.2.3 Air, Water and Foodborne Diseases	**4.2.3.1** Show the ability to prevent the transmission of air, water and food-borne diseases	• Investigating air borne, waterborne and food-borne diseases • Identifying the causative agents (*fungi, bacteria, parasites and viruses*) • Practising ways of preventing airborne, waterborne and foodborne diseases (*reducing smoke, wearing masks, planting trees, safe water handling/storage, hand washing, safe food storage…*)	• Ability to prevent air, water and food-borne diseases demonstrated accordingly

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
4.3 THE ENVIRONMENT	**4.3.1 Environmental Management**	**4.3.1.1** Manage natural resources and waste in the environment	• Describing natural resources and wastes • Exploring natural resources in the local environment (*biotic and abiotic factors*) • Analysing the importance of natural resources (*sources of raw materials for industries, source of fresh air, water and food, source of income, tourism…)* • Devising ways of managing natural resources in the local environment • Identifying types of wastes in the local environment (*liquids and solids* • Devising ways of managing solid and liquid wastes in the local environment (*Three Rs- Reuse, Recycle, Reduce, incinerators…)*	• Natural resources and waste in the environmental managed correctly
4.4 PLANT AND ANIMALS	**4.4.1 Invertebrate Animals**	**4.4.1.1** Classify invertebrates found in the local environment	• Identifying different invertebrates in the local environment through a field trip • Sorting invertebrate animals (*Insects, Worms, Spiders, Snails…)*	• Invertebrates classified correctly
		4.4.1.2 Evaluate the economic importance of invertebrates	• Evaluating the economic importance of invertebrates (*pollination, source of food, decomposition, pest control…*)	• Economic importance of invertebrates evaluated appropriately

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
	4.4.2 Pests and Parasites	**4.4.2.1** Explore ways of controlling pests and parasites in the local environment	• Identifying common pests and parasites in the local environment by collecting the specimen • Distinguishing parasites from pests • Examining the structure and function of the parts of the pests and parasites • Analysing factors which lead to the spread of pests and parasites in the environment • Exploring ways of making pesticides using local materials	• Common pests and parasites in the local environment controlled accordingly
		4.4.2.2 Analyse the economic importance of pests and parasites	• Investigating the economic importance of pests and parasites (*refer to; human health, agriculture, livestock health…)*	• Economic importance of pests and parasites analysed accordingly
	4.4.3 Livestock Production in Zambia	**4.4.3.1** Evaluate the importance of livestock farming in Zambia	• Describing livestock production • Identifying the types of livestock found in the local environment • Evaluating the importance of livestock farming (*source of food, provide employment, income generation, provide raw materials…)* • Exploring ways of improving livestock farming in Zambia	• Importance of livestock farming evaluated satisfactorily

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
	4.4.4 Soil Structure	**4.4.4.1** Analyse soil profile	• Describing soil profile • Profiling soil (*top soil, sub soil, weathered rock and parent rock*) by digging a trench • Drawing and labeling the layers of the soil profile	• Soil profile analysed correctly
		4.4.4.2 Classify types of soil based on their characteristics	• Collecting different soil samples from the local environment • Experimenting on different soil samples to identify the soil types • Deducing the type of soil based on experimental results (*loam, clay and sand)* • Analysing the importance of soil in crop production	• Types of soils classified correctly
	4.4.5 Classification of Plants	**4.4.5.1** Classify plants into flowering and non-flowering plants	• Identifying flowering plants found in the local environment using common names (*monocotyledonous and dicotyledonous*) • Identifying non- flowering plants found in the local environment using common names (*algae, mosses, ferns…*) • Distinguishing flowering plants from non-flowering plants based on their physical characteristics • Investigating the importance of	• Flowering and non-flowering plants classified based on their characteristics accordingly

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
			flowering and non-flowering plants (*ornamental, health, income, source of oxygen …*)	
	4.4.6 Horticulture	**4.4.6.1** Grow common horticultural plants found in the environment	• Describing horticultural plants • Identifying suitable site for growing horticultural crops • Preparing the land for growing horticultural plants • Carrying out simple propagation by (*budding, grafting, cuttings…)* • Evaluating the importance of horticultural plants in the environment (*source of food, source of income, ornamental, medicinal, aesthetic, environmental benefits …)* • Marketing horticultural plants	• Common horticultural plants found in the local environment grown properly
	4.4.7 Crop Production	**4.4.7.1** Grow a variety of field crops	• Selecting suitable site for growing field crops (*legumes, cereals, cucurbits, pumpkin and watermelon*) • Preparing land • Practicing crop management (*seed selection, soil fertility, weed and disease control and harvesting*) • Adding value to the produce • Marketing the crop produce and product	• A variety of field crops grown to maturity successfully

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
4.5 MATERIALS AND ENERGY	**4.5.1 Nature of Matter**	**4.5.1.1** Analyse the nature of matter	• Exploring matter in the environment • Investigating the states of matter (*solid, liquid and gas…)* • Experimenting on the changes of state of matter • Experimenting on the solubility and insolubility of different solid substances • Classifying substances as pure and mixtures	• Nature of matter categorised accordingly
	4.5.2 Types of Materials	**4.5.2.1** Explore different types materials used in the production of items	• Collecting different types of materials from the environment • Determining sources of materials (*mining – metals; forests – timber; fossil fuel, cellulose and salt – plastics*) used in the production of the various items • Exploring the importance of materials (*minerals, timber, and plastics*)	• Different types of materials used in production of items explored accordingly
		4.5.2.2 Categorise materials as metals and non-metals	• Collecting different materials found in the environment • Distinguishing metals from non-metals (*Iron, copper, zinc; wood, plastics, rubber, ceramics…*) • Analysing the physical characteristics of metals and non-metals	• Metals and non-metals categorised correctly

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
	4.5.4 Measuring Matter and Phenomena	**4.5.4.1** Measure quantities of different types of matter and phenomena	• Exploring different types of matter (*solids, liquids gases…)* • Measuring different quantities of matter such as: volume, length and mass in their respective units (*grams, centimetres, cubic centimetres…*) • Measuring different phenomena such as energy and time	• Different types of matter measured correctly
	4.5.5 Nature of Air	**4.5.5.1** Investigate the composition of air	• Experimenting on the composition of air (*water vapour, oxygen, carbon dioxide, nitrogen*) • Designing a pie chart to illustrate the percentage by composition of air using available software	• Composition of air investigated correctly

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
	4.5.6 Simple Machines	**4.5.6.1** Design simple machines	• Describing simple machines • Investigating different simple machines used in everyday activities (*levers, pulleys, wedges, inclined planes, wheel and axle and screw)* • Applying the load, pivot and effort of simple machines • Designing different simple machines using locally available materials	• Simple machines designed appropriately
	4.5.7 Communication Devices	4.5.7.1 Utilise different types of communication devices	• Identifying the different types of communication devices: (*drums, siren, mobile phones, fixed line telephone, television, radio, email/fax, computers, modem ...)* • Using different devices to communicate	• Different types of communication devices utilised satisfactorily
	4.5.8 Light Energy	**4.5.8.1** Explore different sources of light energy	• Describing light energy (*visible light*) • Identifying natural and artificial sources of light (*sun, stars, moon, bulbs...)* • Investigating the uses of light energy (*Photosynthesis, source of electrical energy, growth of human body, drying and evaporation...*)	• Different sources of light explored correctly

GRADE 5

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
5.1 THE HUMAN BODY	**5.1.1 The Skeletal System**	**5.1.1.1** Demonstrate understanding of the skeletal system of the human body	• Describing the skeletal system of the human body • Analysing the skeletal system in man using a model or chart (*Structure- Skull, vertebral column, collarbone, shoulder blades, ribcage, pelvic girdle and limbs. Functions- Support, protection, shape, movement, blood cell production…)* • Examining the functions of the parts of the human skeleton system in man (*structure and function of the bones*) • Comparing and contrasting the human skeleton and other vertebrates • Investigating diseases and disorders affecting the human skeletal system • Making human skeletal models using locally available materials	• Understanding of the skeletal system of the human body demonstrated accordingly
	5.1.2 Puberty	**5.1.2.1** Examine changes associated with puberty in girls and boys	• Describing puberty in boys and girls • Describing physical changes in *girls – enlargement of breast, menstruation, pubic hair, acne, body odour* • Describing physical changes in *boys – deepening of the voice, pubic hair, acne, enlargement of the penis and testes* • Identifying social and emotional changes in both boys and girls	• Changes associated with puberty examined accordingly

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
		5.1.2.2 Practise personal hygiene at puberty stage	• Discussing personal hygiene • Practising personal hygiene by: (*bathing regularly, cleaning armpit and pubic areas regularly, wearing clean clothes*) • Managing menstrual hygiene effectively (*correct disposal of used sanitary pads, regular hand washing, bathing, changing of sanitary pads*) • Demystifying issues around menstruation	• Personal hygiene at puberty stage practiced correctly
		5.1.2.3 Examine risks associated with puberty	• Identifying risks associated with puberty (*teenage pregnancy, STIs including HIV and AIDS, drug and alcohol consumption…)* • Devising ways of reducing risks associated with puberty	• Risks associated with puberty Examined accordingly
5.2 NUTRITION AND HEALTH	**5.2.1 Food Nutrients**	**5.2.1.1** Classify food nutrients according to their function in the body	• Identifying food nutrients according to their function in the body (*proteins, vitamins, carbohydrates, fats and oils, minerals and dietary fibre*) • Carrying out simple food tests on nutrients (*grease spot…)* • Investigating the functions of nutrients in the body	• Food nutrients classified correctly

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
	5.2.2 Labels on Packaged Foods	**5.2.2.1** Interpret the labels on the packaged foods	• Collecting packaged food containers with labels • Identifying nutrients contained and their respective quantities indicated on the information label • Checking manufacturing and expiry dates • Checking quantities to be consumed	• Labels on the packaged foods interpreted correctly
	5.2.3 Malnutrition	**5.2.3.1** Solve triple burdens of malnutrition in the community	• Describing malnutrition in human beings • Investigating prevalence of malnutrition in the community such as under-nutrition, over- nutrition and micro- nutrient deficiency • Practicing eating food in the right groups and quantities (*food portioning*) • Exercising regularly • Taking the Body Mass Index (*BMI)* test regularly • Inviting a nutritionist to discuss the triple burdens of malnutrition	• Triple burdens of malnutrition in the community solved accordingly
	5.2.4 Harmful Substances and their Effects	**5.2.4.1** Investigate the effects of harmful substances on the health of individuals	• Identifying harmful substances that are abused by individuals in the community (*alcohol, drugs such as marijuana and cocaine, solvents…)* • Examining the effects of abused drugs on the body (*aggressiveness, organ damage, restlessness, nausea, addiction, mental illness, isolation, suicide…*) • Exploring ways of preventing substance abuse in the community	• The effects of harmful substances on the health of individuals investigated accordingly

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
	5.2.5 Parasitic Diseases	**5.2.5.1** Explore ways of preventing the spread of parasitic diseases in the community	• Describing parasitic diseases • Investigating common parasitic diseases in the community (*malaria, sleeping sickness, anthrax, acne, yellow fever, toxoplasmosis…)* • Exploring ways of preventing the spread of parasitic diseases (*practicing sleeping under treated mosquito net, keeping the environment clean by slashing tall grass, good hygiene, burying over rubbish pits, residual spraying regularly in the community…*)	• Ways of preventing the spread of parasitic diseases in the community analysed accordingly
	5.2.6 STIs, HIVand AIDS	**5.2.6.1** Analyse ways of preventing infection of STIs, including HIV	• Identifying health risky behaviours associated with the infection of STIs, including HIV infection (*multiple and concurrent sexual partners, unprotected sexual intercourse, sharing unsterilized sharp instruments*, *drug and substance abuse…)* • Practising healthy behaviours to reduce the risk of contracting STIs and HIV (*Regular testing, abstinence from sexual activities….)*	• Ways of preventing STIs including HIV analysed accordingly

TOPIC	SUB -TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
5.3 THE ENVIRONMENT	**5.3.1 Climate Change**	**5.3.1.1** Explore the causes and effects of climate change on the environment	• Describing climate change • Investigating the causes of climate change (*deforestation, indiscriminate burning of bushes, emissions of greenhouse gases…*) • Analysing the effects of climate change (*loss of habitat for wildlife, droughts, hotter temperatures, severe storms, floods…)*	• The causes and effects of climate change explored satisfactorily.
	5.3.2 The Water Cycle	**5.3.2.1** Analyse the water cycle	• Describing the water cycle • Carrying out experiments on processes of evaporation, transpiration, condensation and precipitation • Interpreting effects of water cycles on the environment (*droughts, floods, changes in weather pattern*) • Making models and drawings of the water cycle • Analysing the importance of the water cycle in the environment (*Availability of fresh water, regulating weather patterns...)* • Analysing measures to mitigate the adverse effects in the water cycle (*water conservation and water harvesting, reafforestation, fire guards*)	• Water cycle analysed accordingly

TOPIC	SUB TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
5.4 PLANTS AND ANIMALS	**5.4.1 Soil Health**	5.4.1.1 Explore ways of improving soil health	• Describing soil health • Carrying out experiments to test the fertility of given soil samples • Determining the acidity or alkalinity of the soil • Improving soil fertility (*adding agricultural lime, manure and compost, and inorganic fertilisers*) • Comparing advantage of using organic and inorganic fertilisers • Determining suitable crops for planting in tested soil	• Ways of improving soil health explored accordingly
	5.4.2 Vertebrate Animals	**5.4.2.1** Classify vertebrate animals based on their characteristics	• Classifying vertebrates based on their characteristics	• Different vertebrates classified accordingly
		5.4.2.2 Evaluate the economic importance of vertebrate animals	• Analysing the economic importance of vertebrates (*refer to; Food, Pets, Tourist attraction, Source of raw materials, pollination, vectors of diseases and key players in the ecosystem…*)	• Economic importance of vertebrates evaluated accordingly

TOPIC	SUB TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
	5.4.3 Poultry	5.4.3.1 Demonstrate basic understanding of poultry production	• Describing poultry production • Classifying poultry birds into groups according to their characteristics (*Laying breeds, meat breeds and dual breeds*) • Designing basic housing for poultry • Managing the living environment for birds (*dry and wet cleaning), disinfection, fumigation and feed system*) • Managing the nutrition, health and conditions for the birds (*vaccination, feeding, temperature, ventilation, nutrition, watering*) • Marketing the birds • Adding value to the products	• Basic understanding of poultry production demonstrated accordingly
5.5 MATERIALS AND ENERGY	**5.5.1 Harnessing Electrical Energy**	**5.5.1.1** Explore alternative sources of electrical energy	• Describing electrical energy • Exploring ways of harnessing electrical energy from sources such as wind, solar, hydro, fossil fuels. • Investigating the designs of solar panels in harnessing electrical energy from the sun (*Use locally sourced materials*)	• Alternative sources of electrical energy explored accordingly

TOPIC	SUB TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
	5.5.2 Electricity	**5.5.2.1** Construct electric circuits to demonstrate the flow of electricity	• Describing an electric circuit • Identifying the components of a simple circuit • Constructing a simple circuit that can power a light bulb • Differentiating series and parallel circuits • Exploring the use of various electric appliances • Experimenting with good and bad conductors of electricity • Exploring safety measures when using electricity (*refer to the use of insulators, earthing…*)	• Electric circuits constructed to demonstrate the flow of electricity correctly
		5.5.2.2 Construct a simple Direct Current motor	• Identifying the components of a simple Direct Current motor • Constructing Direct Current motors (*Identifying motors in electrical appliances, collecting necessary materials needed*)	• Simple DC motor constructed correctly

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
	5.5.3 Storage of Energy	**5.5.3.1** Explore ways of storing energy	• Describing the storage of energy and its importance • Exploring different ways of storing energy: (*Compressed air storage, battery storage, liquid air storage…)*	• Ways of storing energy explored correctly
	5.5.4 Separation Techniques	**5.5.4.1** Separate mixtures using an appropriate technique	• Describing different types of mixtures • Analysing appropriate techniques for separating different mixtures (*sorting, decantation, filtration, magnetism, evaporation and sublimation*) • Devising appropriate separating techniques to separate the different mixtures (*sorting, decantation, filtration, magnetism, evaporation, sublimation…)* • Creating a water filter and evaporating different solutions	• Mixtures separated using various techniques appropriately

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
	5.5.5 Simple Machines	5.5.5.1 Construct inclined planes	• Exploring inclined planes and their importance • Applying force to move objects using inclined planes • Identifying inclined planes in everyday life • Designing a simple inclined plane • Using inclined planes in everyday life	• Inclined plane constructed efficiently
	5.5.6 Farm Machinery	5.5.6.1 Use simple farm tools and implements	• Describing farm tools and implements • Making simple farm tools and implements • Using simple farm tools and implements (*hoes, sprayers, ploughs…*)	• Simple farm tools and implements handled and used effectively

GRADE 6

TOPIC	SUB-TOPIC	SPECIFIC COMPETENCES	LEARNING ACTIVITIES	EXPECTED STANDARD
6.1 THE HUMAN BODY	**6.1.1 The Digestive System**	**6.1.1.1** Demonstrate understanding of the digestive system of the human body	• Describing the digestive system in human beings • Making a model of the alimentary canal collaboratively • Drawing and labelling the parts of the alimentary canal • Examining the movement of food in the alimentary canal (*ingestion, digestion, absorption, utilization, egestion*) • Evaluating the importance of the digestive system in human • Investigate the common diseases of the elementary canal (*constipation, ulcers…)*	• Understanding of the digestive system of the human body demonstrated appropriately

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	6.1.2 Risk of Teenage Pregnancy	6.1.2.1 Examine risks associated with teenage pregnancy	• Describing teenage pregnancy • Investigating the causes of teenage pregnancies (*peer pressure, poverty, sexual violence, lack of information…*) • Analysing health, social and emotional risks of teenage pregnancy • Finding solutions to teenage pregnancies (*creating behavioural change in learners to develop positive attitudes towards preventing teenage pregnancies…*) • Creating and sharing simple behavioural change messages with the community members.	• Risks associated with teenage pregnancy examined satisfactorily

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6.2 NUTRITION AND HEALTH	**6.2.1 Health Risks Associated with Nutrition**	6.2.1.1 Relate nutritional disorders to poor nutrition in human beings	• Exploring health risks associated with poor nutrition (*Hypertension, obesity, heart diseases, Marasmus, Kwashiorkor, Rickets…)* • Investigating ways of reducing health risks associated with poor nutrition	• Disorders associated with poor nutrition explored correctly
	6.2.2 Tuberculosis	**6.2.2.1** Practise effective ways of preventing the spread of Tuberculosis in the community	• Aanalysing Tuberculosis (TB) and its causative agent • Identifying common signs and symptoms of TB infection in patients (*basic signs and symptoms*) • Investigating ways of how TB is spread from one person to another (*visit to the nearest health facility-project*) • Identifying persons that are more susceptible to infection of the TB bacteria (*under 5 children, immune-compromised individuals*) • Devising ways of preventing the spread of TB in the community (*improving ventilation, medical checkups, community sensitisations, immunisation and vaccination, contact tracing…)* • Managing and treating of	• Ways of preventing the spread of TB in the community effectively practised

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			persons infected with TB through consistent taking of medication and counselling • Discussing ways of showing empathy with patients of TB in the community (*care, support group….*)	
	6.2.3 Bacteria, Fungi and Viruses	6.2.3.1 Classify bacteria, fungi and viruses based on the structure and mode of infection	• Describing bacteria, fungi and viruses • Identifying similarities and differences of bacteria, fungi and viruses • Draw basic structures of bacteria, fungi and viruses (*observation/pictorial*) • Evaluating the economic importance of bacteria, fungi and viruses (*decomposition of matter, cause diseases, foods, medicines*) • Preventing the spread of bacterial, fungal and viral diseases (*Sensitizing the local community on the spread of the diseases, practicing good hygiene, treatment and vaccination*)	• Bacteria, fungi and viruses classified correctly

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6.3 THE ENVIRONMENT	**6.3.1 Environmental Management**	**6.3.1.1** Develop sustainable practices and approaches in environmental management	• Exploring different ways of managing the environment (*practicing planting trees, and avoid cutting them, avoiding air, land and water pollution, managing waste (solid, liquid commercial, hazardous and e-waste*) • Devising alternative methods of managing waste • Designing and creating a mini green zone on an identified baren land in the school/community (*community gardens/park, pollution free…*)	• Sustainable practices and approaches in environmental management developed satisfactorily
	6.3.2 Water Supply Systems	**6.3.2.1** Examine different water supply systems	• Analysing different water supply systems in the community (*centrally supply system, water reservoirs/river and boreholes, decentralized supply system (wells, boreholes, streams …*) • Investigating different ways of treating water in the community • Designing prototypes of water supply systems.	• Different water supply systems examined appropriately

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		6.3.2.2 Explore water conservation methods in homes and communities	• Describing water conservation and its importance • Investigating ways in which water is wasted in our homes and communities • Practicing ways of conserving water in homes and communities (*turning off a running tap, repairing leaking taps, washing clothes at once, use water efficient appliances, use drought-tolerant landscaping methods…*)	• Water conservation methods in homes and communities explored accordingly

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6.4 PLANTS AND ANIMALS	**6.4.1 Reproduction in Flowering Plants**	**6.4.1.1** Investigate the reproductive parts in flowering plants	• Describing reproduction in flowering plants • Analysing the male and female parts of the flower from samples collected from the environment • Illustrating the parts of the flower • Relating the parts of the flower to its function • Describing the formation of the seed in the flowering plant (*basic description*)	• The reproductive parts in flowering plants investigated correctly
		6.4.1.2 Determine methods of dispersal based on the structure of the seed	• Describing seed dispersal • Exploring the environment to collect different seeds and fruits • Examining the structure of seeds • Deducing the methods of seed dispersal based on the structure of the seed	• Methods of seed dispersal based on the structure determined correctly
		6.4.1.3 Explore different methods of propagating plants found in the environment	• Exploring ways of propagating different plants found in the local environment (*Stems, runners, roots, leaves and bulbs*) • Propagating different types of plants	• Different methods of propagating plants explored

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	6.4.2 Soil Erosion and Conservation	**6.4.2.1** Practice soil conservation	• Describing soil conservation • Practicing soil conservation to reduce erosion (crop rotation, reduced tillage, mulching, cover cropping, cross slope farming, avoiding overgrazing and cutting down trees) • Analysing the importance of conserving the soil • Maintaining a lawn as a project for the class	• Soil conservation practiced accordingly
	6.4.3 Crops	**6.4.3.1** Grow different field crops to maturity	• Exploring different field crops grown in the local environment (*maize, beans, cassava, sunflower, sorghum…*)	• Different field crops grown successfully

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			• Selecting site and Land preparation • Planting of seeds • Managing the crop (*weeding, application of plant nutrients, prevention and control of pests and parasites*) • Harvesting the crop • Storing and marketing of the crop • Exploring other ways of improving the yields	
	6.4.4 Crop Diversification	**6.4.4.1** Grow drought-resistant crops	• Investigating drought-resistant crops (*Some varieties of - millet, cassava, orange, maize, sunflower, soya beans...*) • Selecting site and land preparation • Selecting right seed for the drought-resistance crops • Planting drought-resistance crops • Managing drought-resistance crops • Harvesting and storage • Marketing the drought-resistance crops	• Drought- resistant crops grown to maturity satisfactorily
	6.4.5 Agro-processing and Entrepreneurship	**6.4.5.1** Develop innovations in Agro-processing and entrepreneurship	• Describing Agro-processing • Packaging farm produce into finished products • Branding of finished products	• Innovations in Agro-processing and entrepreneurship developed accordingly

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	6.4.6 Small- Stock Production	**6.4.6.1** Rear small-stock	• Describing small-stock production • Rearing small- stock (*Rabbits, guinea pigs, sheep, goat…)* • Selecting the right breed • Managing small-stocks (*nutrition, health, shelter…)* • Processing products from carcasses for value addition • Marketing the products	• Sall-stock reared to maturity accordingly
6.5 MATERIALS AND ENERGY	**6.5.1 Sustainable Energy Conservation**	**6.5.1.1** Practise efficient energy conserving methods	• Describing conservation of energy • Analysing the challenges faced in sustainable conservation of energy in communities (*inadequate infrastructure, poor policies in place, lack of resources, poor data management, lack of awareness…*) • Devising effective and efficient methods of conserving energy (*switching off lights and appliances when not in use, re-using materials as alternative source for wood to avoid cutting down trees, reducing activities around water recharge areas such as swamps…)*	• Efficient energy conserving methods practised accordingly
	6.5.2 Simple Machines	**6.5.2.1** Use simple machines to do work	• Identifying the load, effort and pivot in simple machines • Classifying levers as first, second, and third levers • Experimenting with levers	• Simple machines used to do work accordingly

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			by altering the positions of the pivot, load, and effort • Designing of levers using local materials	

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	6.5.3 Basic Astronomy	**6.5.3.1** Demonstrate understanding of the relationship between the earth, moon and the sun in the galaxy	• Describing the galaxy • Observing patterns of stars in the sky; our solar system; comets; and the moon • Identifying the source of light in the solar system • Comparing the movement of the earth and the moon in the solar system • Demonstrating the formation of solar and lunar eclipses • Making models of the galaxy	• Understanding of the relationship between the earth, moon and the sun demonstrated accordingly
	6.5.4 Technological Advancement	**6.5.4.1**. Design prototypes of robots	• Describing Robotics • Identifying the different types of robots • Designing a simple circuit of a robot • Describing the function of components of the simple electronic circuit • Designing prototypes of robots to stimulate interest in Robotics	• Prototypes of robots designed successfully
	6.5.5 Sound	**6.5.5.1** Evaluate the sources and properties of sound	• Describing sound energy(*refer to vibrations)* • Identifying various sources of sound *(animals, musical instruments…)* • Anlysing different properties of sound (*pitch, loudness, echo…*) • Analysing the uses of sound energy in everyday life (*communication, warning systems, music…)*	• Sources and properties of sound evaluated accordingly

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	6.5.6 Magnetism	**6.5.6.1** Apply the principles of magnetism	• Describing types of magnets • Exploring the laws of magnetism (*repulsion and attraction*) • Identifying the poles of a magnet • Exploring the uses of magnets in everyday life (*compass, speakers, electric motors, electric trains…)*	• The principle of magnetism applied accordingly

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